

token: A Digital OPD Token Queue Management SaaS

A Semester Project for Basic Toolkit in Data Science

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Abstract

The Outpatient Department (OPD) in Indian government hospitals frequently experiences severe overcrowding and systemic strain, leading to multi-hour wait times for patients. This case study details the development of a Minimum Viable Product (MVP) designed to address a critical symptom of this problem at Goa Medical College & Hospital (GMC): the physical, patient-managed queue that begins at 6 AM, three hours before the counters officially open. The MVP's primary goal was to replace this analog "placeholder" queue with a simple, accessible digital token system. This report outlines the background, the specific problem, the focused MVP solution, the technology stack, and the success metrics defined for this strategic intervention.

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1 Background: The OPD Crisis in Public Healthcare

1.1 A System Under Strain

The Outpatient Department (OPD) is the "shop window" of any hospital, serving as the primary interface between the public and the healthcare system. In government hospitals across India, this interface is defined by systemic strain, severe overcrowding, and unstructured patient flow. A single large hospital OPD can see over 1,200 patients daily, a number that doubles when accounting for patient attendants.

This massive influx overwhelms the prevalent manual, paper-based systems. Empirical data reveals that 70% of patients wait over two hours just for registration, with 63.3% waiting another hour for their consultation. This inefficiency is a fundamental failure in service delivery, creating a perpetually stressed environment.

1.2 The Human Cost

The operational metrics of long queues translate into a significant human cost. For patients, the journey is marked by frustration, anxiety, and physical exhaustion, particularly for the elderly and chronically ill. This negative experience is the primary driver of dissatisfaction with public healthcare.

For hospital staff, the environment is equally taxing. The constant pressure of managing disorderly crowds contributes to significant stress and burnout. Doctors and nurses, whose primary function is clinical care, must divert attention to crowd management and administrative tasks, reducing the quality of care for everyone.

2 The Specific Problem at Goa Medical College & Hospital (GMC)

While GMC's OPD is semi-digital (using a system for case paper registration), it suffers from a unique and acute problem rooted in its "first come, first serve" policy.

- **Counter Timings:** OPD counters officially open at 9:00 AM.
- **The Real Queue:** To secure a spot, patients and their families begin to arrive and form a queue as early as 6:00 AM.
- **The Analog "Placeholder" System:** Patients use personal items—such as shopping bags, medical documents, or even footwear—to "reserve" their place in the line for three hours.
- **The Core Issue:** This analog system forces patients and their families, many of whom are ill and have traveled long distances, to wake up exceptionally early and endure a long, uncomfortable, and unnecessary wait just to be present for a queue that hasn't officially started. This is the specific, high-impact problem the MVP was designed to solve.

3 Proposed Solution: token

The goal of token was not to solve the entire OPD crisis, but to surgically target the 6 AM queue problem. The solution was a lightweight, web-based digital token system with three core components.

3.1 Component 1: Patient Portal

A simple, mobile-friendly website accessible to patients from 6:00 AM onwards.

- **Function:** Allows a patient to enter their existing Case Paper ID and select a department.
- **Output:** Generates a digital token number (e.g., G-001).
- **Impact:** Directly replaces the need for a physical placeholder. A patient can now secure their token from home at 6 AM and arrive at the hospital closer to 9 AM.

3.2 Component 2: Counter Console

A secure web dashboard for the OPD counter staff.

- **Function 1:** Displays a real-time, chronologically-ordered list of all tokens generated from the Patient Web Portal.
- **Function 2:** Allows staff to issue new tokens for walk-in patients (those without a smart-phone or digital token), seamlessly merging them into the same digital queue.
- **Function 3:** Provides tools for staff to categorize tokens as "General," "Senior Citizen," or "Referral," enabling them to manage the queue flow.

3.3 Component 3: Admin Dashboard

A secure, read-only web portal for hospital administrators.

- **Function:** Displays high-level, real-time analytics, such as the total number of tokens issued and the breakdown by category.
- **Impact:** Provides basic operational oversight without complicating the MVP.

4 Technology Stack and Implementation

The was built using a modern, scalable, and cost-effective technology stack designed for rapid development and real-time data synchronization.

4.1 Technology Stack

4.2 Implementation Process

The development followed a streamlined process:

Table 1: MVP Technology Stack

Component	Technology	Purpose
Frontend	React.js	Building the user interfaces for all three web-based components (Patient, Counter, Admin).
Backend	Firebase (BaaS)	Providing a "Backend-as-a-Service" to eliminate the need for server management.
Database	Firestore	A NoSQL, real-time database used to store and instantly sync token data across all three consoles.
Authentication	Firebase Auth	Securing the Counter and Admin dashboards with email/password login for staff.
Hosting	Firebase Hosting	Deploying the React web application globally with a single command, providing SSL and CDN by default.

1. **Firestore Setup:** A new project was created in the Firebase console. Firestore database and Email/Password Authentication were enabled.
2. **React App Creation:** A standard React project was initialized using 'create-react-app'. The Firebase SDK and React Router were installed.
3. **Component Development:** The three core components (Patient Portal, Counter Console, Admin Dashboard) were built as separate routes in the React application.
4. **Real-time Logic:** The Counter Console used Firestore's 'onSnapshot' listener to get a live feed of new tokens as they were generated by patients.
5. **Deployment:** The production build of the React app was deployed using the Firebase CLI 'firebase deploy' command, making it instantly available online.

5 Conclusion and Future Work

5.1 Conclusion

This case study demonstrates a strategic approach to digital transformation in a resource-constrained public health environment. By identifying the most acute patient pain point—the 6 AM physical queue—and developing a highly focused MVP, the project aimed to deliver significant and immediate value with minimal complexity. It replaces a system of "anxiety-driven waiting" with a digital-first system of "at-home convenience," laying a critical foundation for broader digitization efforts.

5.2 Future Roadmap

The MVP intentionally deferred several features to ensure rapid deployment. The success of this MVP would justify a Phase 2, which would include:

- **Public Display Screens:** Integrating digital displays in the waiting halls to show token numbers.
- **Doctor's Console:** A simple interface for doctors to "call" the next patient, automating the flow from counter to consultation.
- **Advanced Sorting:** Implementing the hospital's complex sorting logic (for Senior Citizens, Referrals, etc.) as an automated algorithm.
- **Notifications:** Adding SMS or WhatsApp notifications to alert patients when their turn is approaching.